Amogha Padmanabha

Embedded Systems Engineer

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Website Link Linkedin Link 26.08.2001

Skills

Hardware & Digital Design
Verilog, VHDL, RISC-V, Processor Design, PCB Design

Systems & Low-Level Software
C++, Rust, Embedded Linux, Yocto, System Integration

Firmware & Embedded Systems
Embedded C, Driver Development, RTOS, SPI, I2C, UART, BLE, WiFi.

Application & Software Engineering
Python, Modular Software Design, API Development, Testing & Validation, Git, Jira, CI/CD

Professional Experience

Master Thesis, *Belden* — Oct 2025 – Present, Stuttgart,Germany

- Designed and implemented a stateful Fault Detection, Isolation, and Recovery framework for service-oriented embedded systems
- Developed runtime fault episode tracking and dependency-aware recovery coordination
- Ensured deterministic execution for reliable system behavior
- Optimized for low CPU and memory overhead

R&D Intern, *Belden* — Feb 2025 – Sep 2025, Stuttgart,Germany

- Developed a Rust-based proxy application
- Implemented Rust-C++ interoperability, enabling integration
- Integrated applications into Yocto-based embedded Linux systems
- Collaborated in Agile teams (Git, Jira), improving development efficiency

Research Assistant, *Universität Stuttgart* — Jan 2024 – Jan 2025, Stuttgart,Germany

- Developed BLE firmware in Embedded C for ARM Cortex-M microcontroller
- Implemented communication using SPI, DMA.
- Designed and validated custom PCBs
- Implemented Over the Air Update via BLE

Projects

Dual-Core RISC-V Processor Design

- Designed and implemented a dual-core RISC-V processor using Verilog
- Integrated cache and interrupt handling mechanisms for efficient execution
- Deployed and validated design on Xilinx Spartan FPGA platform

Personal Portfolio Website

- Deployed and maintained live at: https://amoghamp.site
- Built and deployed a responsive personal portfolio using React, Tailwind CSS, and Vite, hosted on Vercel.

Skin Cancer Detection System

- Developed a deep learning-based melanoma detection system using EfficientNet-B7
- Deployed and optimized model on Raspberry Pi for low-cost embedded inference
- Built as part of a government-funded project, focusing on accessible healthcare solutions

Education

Masters in Information Technology, *University of Stuttgart* Oct 2023 – Present
Germany

Bachelors of Engineering in Electronics and Communication, Aug 2019 – May 2023
Visvesvaraya Technological University India

Awards

Best Student Achiever Award, *Visvesvaraya Technological University*

International Appathon, *TiE Global*

Innovative Mini-project Competitions, *Ministry of Education Awards*

Languages

English
C1

German
A2